NATIONAL INSTITUTE OF ENGINEERING

END SEMESTER EXAMINATION - MAY 2027

Duration: 3 Hours Max Marks: 80

Instructions: Attempt any four questions. Figures to the right indicate full marks.

Q1(a) Describe the scheduling policy and derive its average waiting time. [5]

Q1(b) Differentiate between the two normalisation forms with a worked example. [5]

Q1(c) Discuss the error control mechanism and analyse its overhead. [5]

Q1(d) Derive the expression for throughput and state all assumptions clearly. [5]

Q1(e) Compare the two indexing structures and evaluate their lookup complexity. [5]

Q2(a) Differentiate between the two normalisation forms with a worked example. [5]

Q2(b) Discuss the error control mechanism and analyse its overhead. [5]

Q3(a) Discuss the error control mechanism and analyse its overhead. [5]

Q3(b) Derive the expression for throughput and state all assumptions clearly. [5]

Q3(c) Compare the two indexing structures and evaluate their lookup complexity. [5]

Q3(d) Explain the deadlock avoidance algorithm and apply it to a sample matrix. [5]

Q4(a) Derive the expression for throughput and state all assumptions clearly. [5]

Q4(b) Compare the two indexing structures and evaluate their lookup complexity. [5]

Q4(c) Explain the deadlock avoidance algorithm and apply it to a sample matrix. [5]

Q4(d) Illustrate the encoding scheme and compute the resulting bandwidth. [5]

Q4(e) Analyse the concurrency protocol and prove its serialisability. [5]

Q4(f) Describe the routing metric and calculate the shortest path for the graph. [5]

Q5(a) Compare the two indexing structures and evaluate their lookup complexity. [5]

Q5(b) Explain the deadlock avoidance algorithm and apply it to a sample matrix. [5]

Q5(c) Illustrate the encoding scheme and compute the resulting bandwidth. [5]

Q6(a) Explain the deadlock avoidance algorithm and apply it to a sample matrix. [5]

Q6(b) Illustrate the encoding scheme and compute the resulting bandwidth. [5]

Q6(c) Analyse the concurrency protocol and prove its serialisability. [5]

Q6(d) Describe the routing metric and calculate the shortest path for the graph. [5]

Q6(e) Explain the layered reference model and justify each layer with a diagram. [5]

Q7(a) Illustrate the encoding scheme and compute the resulting bandwidth. [5]

Q7(b) Analyse the concurrency protocol and prove its serialisability. [5]

Q8(a) Analyse the concurrency protocol and prove its serialisability. [5]

Q8(b) Describe the routing metric and calculate the shortest path for the graph. [5]

Q8(c) Explain the layered reference model and justify each layer with a diagram. [5]

Q8(d) Describe the scheduling policy and derive its average waiting time. [5]